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DATA-470 Showcase Reference Pack

Backorder risk, model evidence, and plain-English talk tracks

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30-Second Story opening

I built an early warning system that predicts which SAP order lines are most likely to become backorders, using only information available when the order is placed.

Problem: backorders delay customers; overstock ties up working capital. The goal is not to magically fix inventory, but to help planners know which orders deserve attention first.

Project fact	Plain-English meaning
31,177 order lines	The model learned from a meaningful order history.
3.38% positive rate	Backorders are rare, so accuracy alone is not enough.
6,521 temporal test rows	Final check used later orders the model had not trained on.
58 test positives	Small positive count makes the evaluation strict and sometimes unstable.

Headline Result metrics

Model	Role	Temporal holdout result	How to explain it
Logistic Regression	Deployable recommendation	ROC-AUC 0.910; PR-AUC 0.358; precision 0.453; recall 0.586; F1 0.511; threshold 0.309	Simpler model, strongest practical result, easier to defend.
OOF-calibrated stack	Rule-selected champion	ROC-AUC 0.936; PR-AUC 0.326; precision 0.410; recall 0.431; F1 0.420	Won the inner selection process, but did not beat LR on the final future-facing test.

Model gate passed precision ≥ 0.15 and recall ≥ 0.35 . Deployment gate blocked newest labels are not mature enough yet.

Flow and Leakage Guardrails

[plain english](#)

Step	What happened	Why it matters
1	Raw SAP CSVs became cleaned sales, delivery, billing, inventory, purchasing, and master-data tables.	Creates a repeatable pipeline instead of one-off spreadsheet work.
2	Built the v2 order-time modeling table.	Only uses facts that would be known when the order is placed.
3	Trained on older orders and tested on newer orders.	Matches the real question: can past history predict future backorders?
4	Removed future-looking columns like outstanding quantity, delivery status, billing status, and backorder amount.	Prevents the model from getting answers it would not have in real life.

Randomly mixing old and new orders is like studying with some of the test answers accidentally included. It can make the model look smarter than it really is.

Model Definitions

[analogies](#)

Model	Short definition	Analogy
Logistic Regression	Simple model that weights each signal.	A checklist where each item adds or subtracts risk.
Random Forest	Many decision trees voting together.	A panel of judges taking the majority view.
XGBoost / LightGBM	Tree models that learn from earlier mistakes.	Each new reviewer focuses on what the last reviewer missed.
kNN	Predicts from similar past examples.	"Orders like this one usually behaved this way."
OOF-calibrated stack	Combines several models using honest out-of-fold predictions.	A coach trusts each specialist only after seeing how they handled unfamiliar cases.

Edge Cases and Weak Spots gotchas

Issue	Simple explanation	Analogy
Rare positives	Backorders are uncommon, so the model must be judged on how well it catches rare events.	A fire alarm is judged on rare fires, not ordinary quiet days.
Label maturity	Some newer orders may not have had enough time to reveal the true outcome.	Grading a test before every student has turned it in.
Recent holdout collapse	A very small number of recent backorders can make the score swing dramatically.	Judging a fire alarm during a week with almost no fires.
Distribution drift	Older business patterns may not match newer order behavior.	Practicing against last year's team, then facing this year's team.
Rolling features hurt	Some added history features inherited incomplete labels and changing patterns.	Building a weather forecast from a thermometer that stopped reporting reliably.

Likely Questions answers

Question	Answer you can say out loud
Why not use accuracy?	Because backorders are rare. A model could say "no backorder" almost every time and look accurate while missing the useful cases.
Why temporal split?	In real life, we use past orders to predict future orders. Testing older-to-newer is more honest than mixing time periods.
Why LR if stack was selected?	The stack won the formal selection process, but logistic regression was simpler and stronger on the final future-facing test.
Why did feature engineering hurt?	The added features depended on history that was incomplete or changing. More features are not always better if the source signal is shaky.
Is it production-ready?	The model clears basic performance checks, but I would wait for better label maturity before deploying it.

How To Explain It By Audience talk tracks

Audience	Talk track
Non-technical visitor	"Think of this like a watchlist. It helps planners decide what to inspect first."
Business visitor	"The tradeoff is missed backorders versus false alarms. A stricter alarm wastes less time; a more sensitive alarm catches more risk."
Technical visitor	"I tested older orders against newer orders, removed future-looking fields, and used rare-event metrics because backorders are uncommon."
If you get stuck	"The key finding is that the model is promising, but data maturity determines whether it is deployable."

Future Plans roadmap

Short version: the next lift probably comes from better business signals, not just fancier algorithms.

- **Add better order-time signals:** available-to-promise status, inventory buffer, supplier/material lead time, plant capacity, purchase order timing, and customer/material history.
- **Score at more than one moment:** keep the order-time model, then add a second checkpoint closer to ship date when more inventory and confirmation data is available.
- **Use fresher training windows:** train on recent years instead of decades if business patterns have changed.
- **Split the problem into smaller groups:** build separate models by plant, material family, or customer segment when there are enough backorder examples.
- **Change the model structure if needed:** move from one universal classifier to a staged system: first flag possible risk, then estimate urgency, then recommend planner action.
- **Improve labels first:** wait for outcomes to mature before judging the newest orders, so the model learns from a complete answer key.

Quick Glossary

[terms](#)

Term	Brief definition	Analogy
Accuracy	Percent of total predictions correct.	A quiz score, but misleading if almost every answer is "no backorder."
Precision	When the model says "risk," how often it is right.	How many alarms were real fires.
Recall	How many real backorders the model caught.	How many actual fires the alarm detected.
F1	Balance between precision and recall.	One score for being careful and not missing too much.
PR-AUC	Ranking quality for rare events.	How well risky orders rise to the top of the watchlist.
ROC-AUC	Overall ability to rank risky above non-risky orders.	Sorting suspicious packages ahead of normal ones.
Threshold	Cutoff for calling an order risky.	How sensitive you set a smoke alarm.
OOF	Predictions from rows a model did not train on.	Practice answers graded by a teacher who did not write them.
Leakage	Using information that would not be known yet.	Studying with some test answers included.
Label maturity	Enough time has passed to know the true outcome.	Grading only after all tests are turned in.

Hard Question Anchor

[fallback](#)

The capstone value is that it separates model quality from data readiness. The model is promising, but the deployment decision depends on whether the newest outcomes are complete enough to trust.